

NANA STUDIOS

• THE NANA • CARNIVAL



Featuring

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CARNIVAL MENU

NANA SPECIALITIES

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PROJECT OVERVIEW



Nana Carnival is a VR game that allows users to play in a virtual carnival that tracks their scores in real-time and is also designed with inclusivity in mind. This game will be created within Unity and will require a VR headset to run.

Goals & Objectives

The main idea we have in mind is to create a game that not only caters to the general public, but to as many people possible. To do this, we would have to implement several design changes to our original game idea to make it accessible to those who need it.

Goals

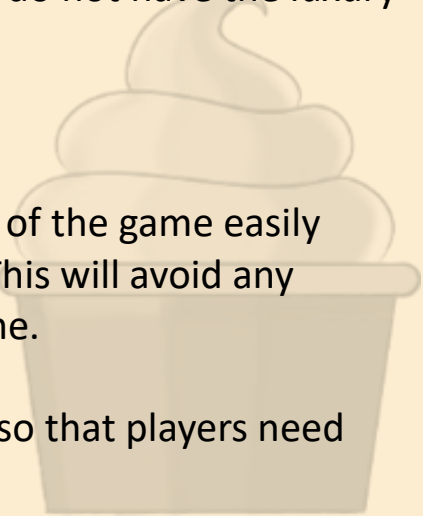
Our goal is to create an immersive VR game that caters to those who are affected by Protanopia and/or Deuteranopia colour blindness, so that they can still experience the game to the fullest.

We also aim to accommodate to those who do not have the luxury to play in a large space.

Objectives

Our objective is to make all features/details of the game easily distinguishable by using colours mindfully. This will avoid any misconceptions/difficulty in playing the game.

We also intend to implement teleportation so that players need not walk around physically.



Design Rationale



Game

Due to the current global situation, many wide scale events/activities have been put to a halt for roughly 2 years. This includes leisure entertainment such as concerts, carnivals and conventions. Our team saw an opportunity in this and wanted to create a VR game that will be able to bring one of such widely missed events back into our daily lives while also remaining responsible.

We decided on a carnival with a variety of mini games. Within the fairground, every feature is exposed and there is no need for exploration to unlock any new features. Teleportation is also made use of in the game. Doing so reduces the amount of physical space required in the real world, removing the limitation of limited space since players will not need to walk around physically.

All mini games implemented are simple real life carnival games incorporated into a 3D virtual environment. This makes players feel a sense of familiarity even if it is their first time playing a VR game. All games require similar actions to be made within the game as in real life.

Additionally, the environment is fully built to immerse the players better in the game, making them feel like they are physically at a carnival. Our 3D models, however, are rather low poly to remain economical.

Lastly, we also implemented a leader board in each mini game to fuel the player's competitiveness.

Design Rationale



Inclusivity

Throughout the designing process of our game, we were extremely mindful of the colours used in order to ensure that all features/objects were easily distinguishable for those who have Protanopia and/or Deuteranopia colour blindness. To do so, we tested out the visibility of our colour scheme on Adobe Photoshop through the use of the in-built colour blind filters. Through this process, we were able to pick out suitable colours that were easily distinguishable and also looked aesthetically pleasing. In our game, we made sure the colors that were more important, stood out more compared to the background. We wanted to make sure it was eye-catching and that they would be able to tell apart the important things better.

In addition to being mindful of the colours used, we have also incorporated symbols/icons wherever possible and used as little text as possible. This was done in hopes that international and/or intellectually disabled players will have an easier time navigating our game due to either language barrier or learning disability.

Within the game, all necessary functions are easily accessible. For example, the ticket machine, which is frequently interacted with, is placed in the center of the playing field so that it is of equal distance from each game booth. This also reduces confusion on where the ticket machine may be if forgotten.

In regards to our models, we made slight modifications in certain portions so that the game can be played while seated down or standing up. This is to accommodate players who are physically disabled waist down and/or players who have a very limited playing area.

Competitor Analysis



Upon researching the VR game market, we have found that there are a few games with a similar concept as ours.

RADtv :

Competitor	App Name	Rating	Mini Games	Accessibility	Leader Board
Rockstar Dundee	RadTV	7.5/10	✓	✗	✓
Nana Studios	Nana Carnival	NA	✓	✓	✓

Although RADtv has most of the features that Nana Carnival provides, it is still not accessible to minority groups such as the physically disabled and/or the colour blind. Though it can be played seated, it is a very fast paced game which needs to be recalibrated during each player's turn. The intense gameplay often causes massive fatigue and the recalibration timing affects the player's moods due to long wait times.

Carnival Games:

Competitor	App Name	Rating	Mini Games	Accessibility	Leader Board
Cat Daddy Games	Carnival Games	4/5	✓	✗	✓
Nana Studios	Nana Carnival	NA	✓	✓	✓

Carnival Games® VR is an old VR carnival game that has 12 different mini games and it includes a leader board linked to Steam. However, the downside is that it is inaccessible to various group, mainly the disabled and the colour blind. Though it has variety, it has long been abandoned by it's developers and the game can no longer be run smoothly.

Research

To design a game that is optimized and suitable for VR, our team did some research to ensure that players receive the best experience we can possibly provide.

Hardware (VR Headset)

VR may be immersive but the VR headset comes with several cons. Though the VR headset comes with lenses to aid with myopia, those with severe myopia are still required to wear glasses/contact lenses to view the game clearly. To cater to this, we made sure that our texts were clear and large so that players do not need to strain their eyes to read.

Other limitations of the headset include the length of cable of the tethered VR headset, the heat generated from the headset and the fatigue caused by the VR headset due to extended use.

Target Device

One thing that we had to consider hardware wise was the experience that users would have while playing on different VR headsets. We wanted to make sure that our game was fully compatible with both types of VR headsets; i.e. tethered/untethered.

To make it compatible with the tethered headset, we made sure that players could teleport in the game and play while seated/standing in one spot, reducing the chances of the player tripping or getting tangled in the wires.

To make the game compatible with the untethered VR headset, we made sure to use low poly models to ensure that the game could run smoothly. The full game also does not require a long time to complete. This means that the battery life of the VR headset is unlikely to affect gameplay.

Technical Requirements

To run our game properly, there are several conditions that need to be met.

Hardware

To run our game, players are required to have a VR headset and a system with the following requirements minimally.

GPU

Integrated Intel HD Graphics 620
NVIDIA GeForce MX150 or greater
NVIDIA GeForce GTX 965M or greater
AMD Radeon RX 460/560

Video Output

HDMI 1.4 or DisplayPort 1.2

CPU

Intel Core i5-7200U dual-core

USB Port

USB 3.1 Gen 1

RAM

8 GB DDR3 or more

Bluetooth

4.0

Storage

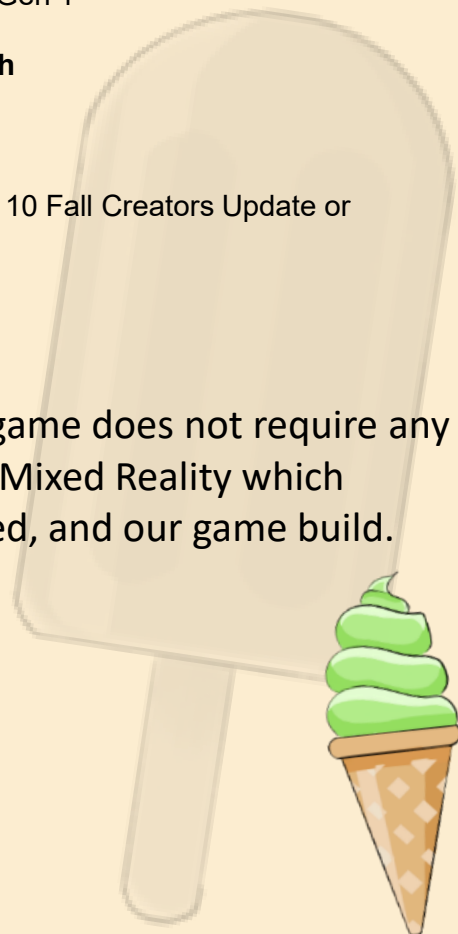
10 GB

OS

Windows 10 Fall Creators Update or later

Software

Regarding software requirements, our game does not require any external software other than Windows Mixed Reality which should, by default, already been installed, and our game build.



Inclusive Design

Inclusive design is a design process that we will take to ensure that our game is playable by as many different groups of people as possible. This means including not only the general public but also tweaking our game to make it accessible to other minority groups.

Who are we catering for?



The main group of people we are catering to are those suffering from Protanopia and/or Deuteranopia colour blindness. The other group of people would be those who do not have a big space to play the game.



Are there any groups excluded?

For our game, we are unfortunately unable to cater to those who are unable to make use of their arms/hands as we are unable to modify the VR headset.



Considerations for Inclusive Design

We have chosen to be mindful of the use of colours for our game.

Firstly, we carefully picked the main colours to ensure that the colours would still look good for everyone. Secondly, we tried to minimise the use of too many different colours by making use of icons/symbols. Doing so allows those who have colour blindness distinguish between objects easily.

SUPPORTING VISUALS



To prepare for the development of our game, our team brainstormed and created visuals in various forms. Doing this aided us in creating a game with consistent style and feel. Additionally, it gave us a clear path of what was necessary for our game to be a success.

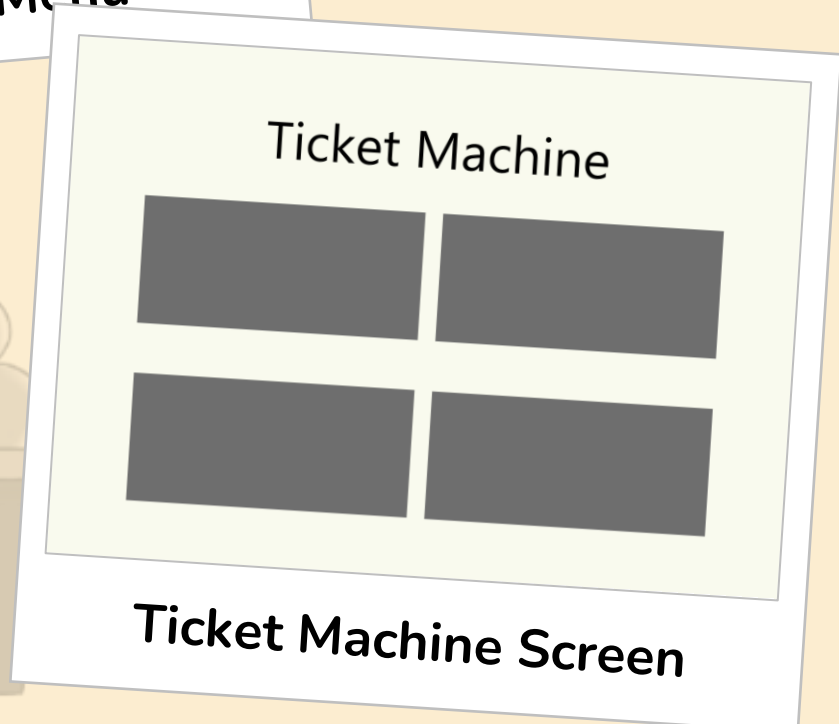


Wireframes



Low Fidelity Wireframe for UI:

To help with the designing process of the game's UI, we made low-fidelity designs on Adobe XD. This is so that we would be able to better visualise better where they should go.



Wireframes



Back

Logo

Registration Form

Email :
Username :
Password :
Confirm Password :

(confirmation text here)
(warning text here)

Register

Register

Logo

Login Form

Email :
Password :

(confirmation text here)
(warning text here)

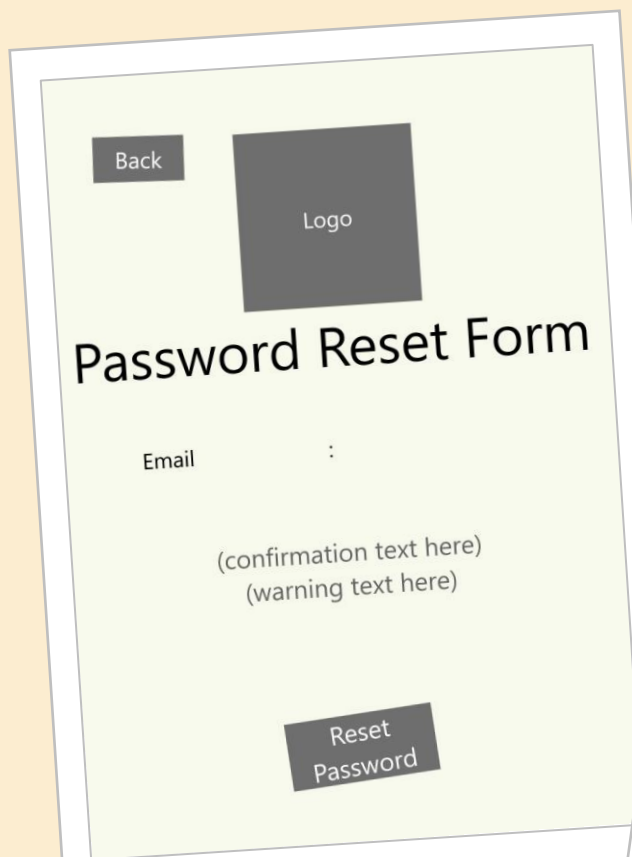
Login

Reset Password Register

Login



Wireframes



A wireframe for a password reset form. It features a 'Back' button in the top left, a 'Logo' placeholder in the top center, and the title 'Password Reset Form' in the center. Below the title is an 'Email' label followed by a colon and a text input field. Underneath the input field are two lines of placeholder text: '(confirmation text here)' and '(warning text here)'. At the bottom right is a 'Reset Password' button.

Back

Logo

Password Reset Form

Email :

(confirmation text here)
(warning text here)

Reset Password

Password Reset



A wireframe for a game booth tutorial screen. It has two 'Icon' placeholders at the top corners and the title 'Tutorial' in the top center. Below the title is a large rectangular placeholder labeled 'Pic from game'. Underneath this is the text 'How to play:' followed by another large rectangular placeholder labeled 'Info about game'.

Icon

Tutorial

Icon

Pic from game

How to play:

Info about game

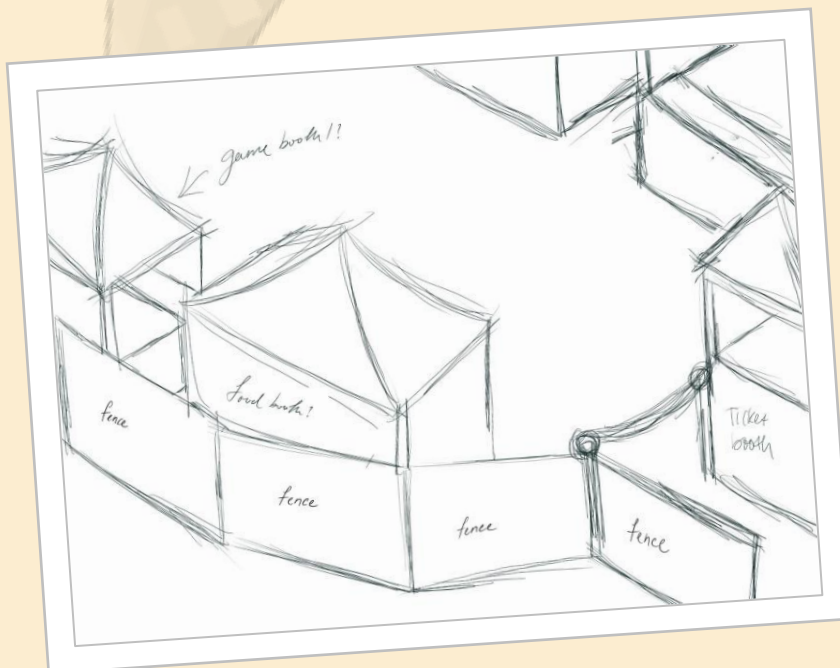
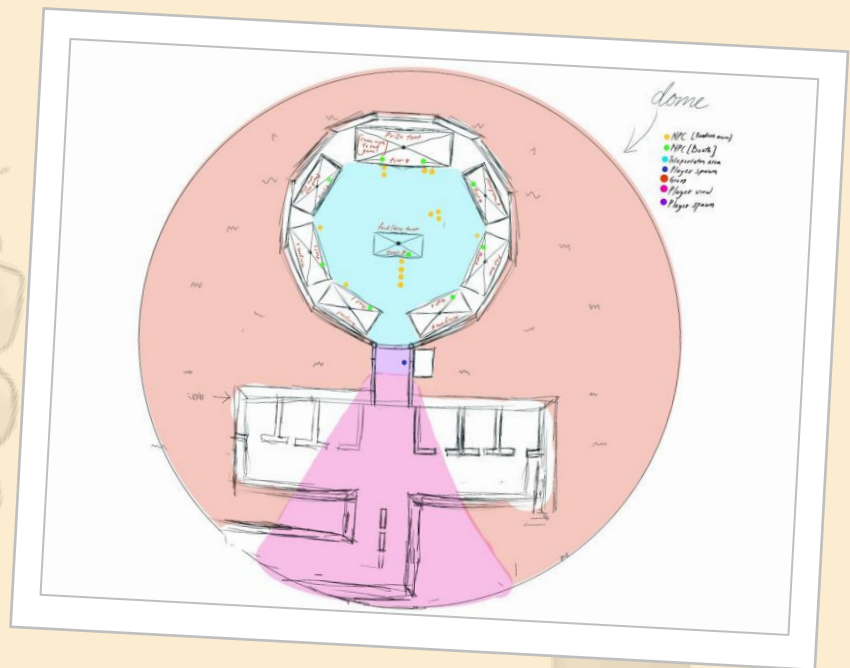
Game Booth Tutorials

Sketches

Sketch of Game Scene Layout:

In order to help with the level building for our game, we came up with a few sketches to give a brief imagery description of what the world will be looking like when we start building.

Top View



3/4 View

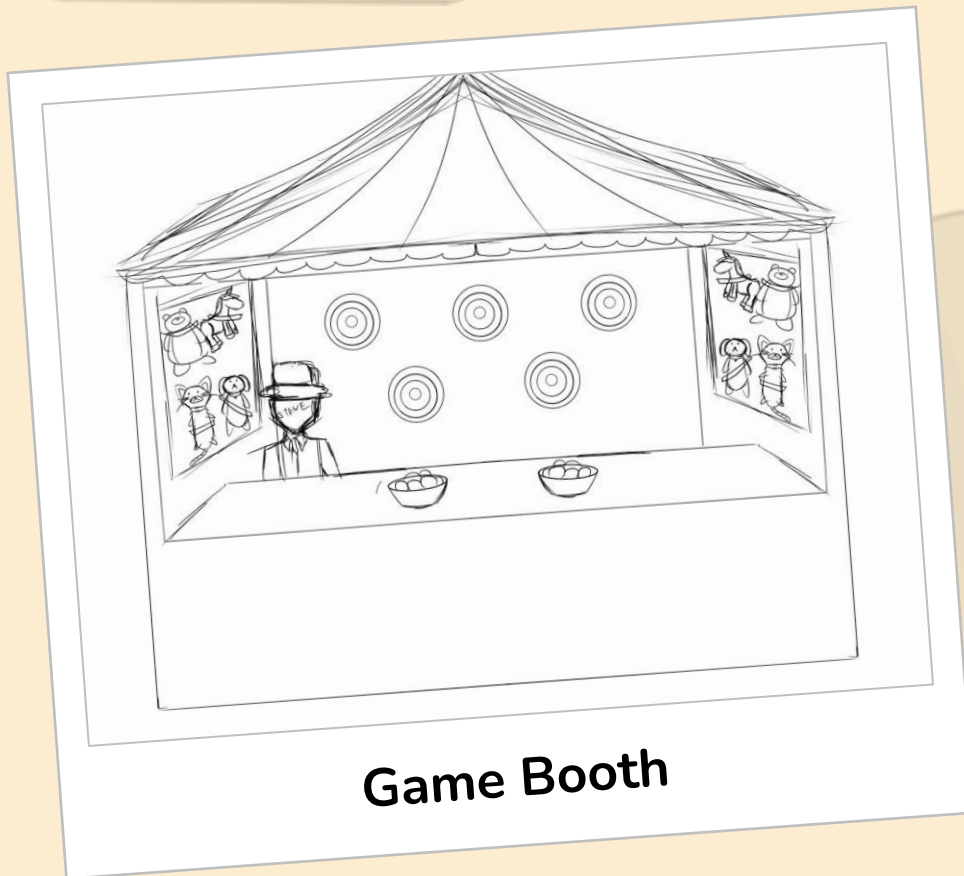
Sketches

Others:

Other than the layout of the game environment, we also sketched up other drafts.



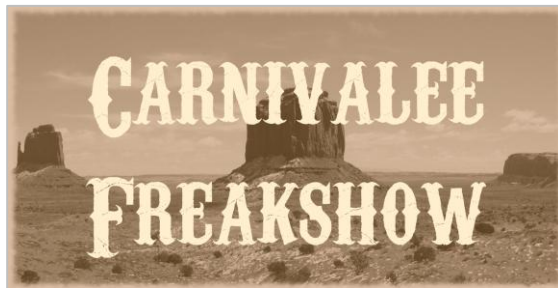
Game Mascot



Game Booth

Mood Board

We created a collage of images, fonts, and colour schemes that we wanted to possibly apply to our game. The purpose of the mood board was to ensure we were going in the right artistic direction before starting on creating the assets needed.



Photoshop Mockups

The following are photoshop mockups created by using various images that have been put together to give help us during the building and designing process.



Photoshop Mockups



Game Booth



Lock Icon

USER PERSONAS



As a team, we decided to interview 2 people from different groups.

Jotham's main role is to act as the control for the game testing to make sure that the game can be played smoothly and if the colours and designs suit the theme.



Lucas Lum on the other hand has colour blindness and will test our game to let us know if everything is clear and distinguishable, whether the colours are suitable for those suffering from Protanopia/Deutanopia.



By cross referencing their responses, we were able to make amendments accordingly that gave us the best of both worlds.





OUR CHEFS

@Nana Carnival



Name: Jotham Wong Status: In a relationship

Nationality: Singaporean Sex: Male

Archetype: Sentinels Age: 19

Salary: NA Tier: NA

1. Bio

Jotham is a year 2 student in Singapore Polytechnic. He is currently pursuing Biomedical Science. He is a hardworking student that has high focus in everything he does. He is the type of person who will accomplish whatever he sets his mind on. He is also highly competitive about everything as he always strives to be the best.

During his leisure time, Jotham often indulges in games. He has always wanted to try out VR games, however, he does not have the space needed to play said games.

2. Personality

- Ambitious
- Optimistic
- Hard-working
- Out-going
- Competitive

3. Frustrations

- Unable to find fun VR games that don't require physical movement
- Unable to find a game with a variety of activities

4. Goals

- To find a fun VR game that reduces his motion sickness
- To find a game with non repetitive gameplay

5. Motivations

Design: 65%
Competition: 100%
Community: 85%
Variety: 90%

6. Behaviors

Usage (features): 85%
Aesthetics: 60%
Simplicity (interface): 95%
Intractability: 70%

7. Frequently Used Apps



Beat Saber



Subnautica



The Climb



OUR CHEFS

@Nana Carnival



Name: Lucas Lum Status: Single

Nationality: Singaporean Sex: Male

Archetype: Entrepreneur Age: 18

Salary: NA Tier: NA

1. Bio

Lucas is currently a year 2 student in Ngee Ann Polytechnic pursuing a Diploma in Electrical Computer Engineering. Although often easily distracted, he will strive to achieve his goals when determined.

Lucas enjoys playing a variety of games from different genres. The ones he plays most are shooter and simulation games. However, due to his colour deficiency, when he plays fast-paced games like Titanfall 2, the colour of the enemy will appear duller. This sometimes makes him miss enemies unknowingly.

2. Personality

- Happy-go-lucky
- Hard-working
- Optimistic
- Active

3. Frustrations

- Missing enemies during fast-paced games due to colors being too dull
- Clunky controls that do not feel good to use

4. Goals

- Able to play a game with colours that will not affect much
- To find a competitive game with controls that are intuitive and easy

5. Motivations

Art style: 35%
Gameplay: 40%
Storyline: 15%
Competition: 10%

6. Behaviors

Usage (features): 50%
Aesthetics: 30%
Simplicity (interface): 10%
Intractability: 30%

7. Frequently Used Apps



Ace
Combat
7



Elite
Dangerous

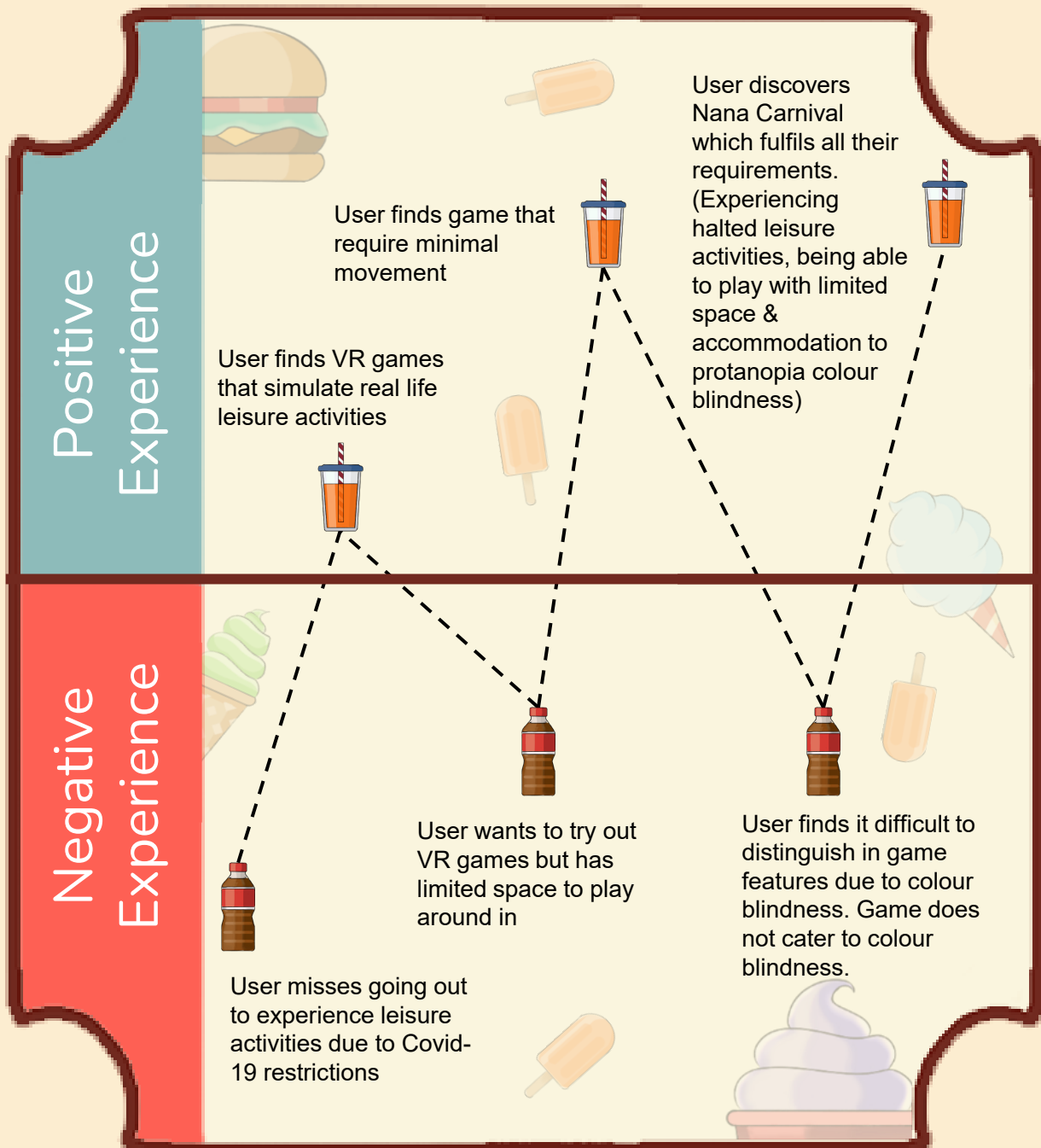


Onward

USER JOURNEY



The flow of events from the point of view of a consumer.



USABILITY TESTING



Usability Testing:

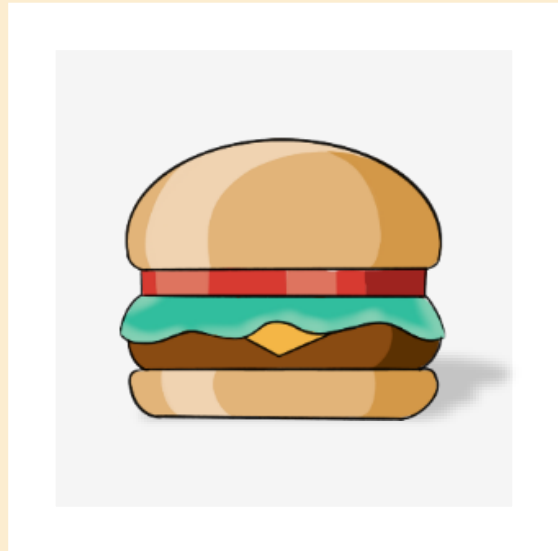
Link:

https://ivid2.np.edu.sg/media/t/1_da2ws5ue

Usability testing was done with Lucas Lum who has protanopia colour blindness.

Details changed after testing:

1. Fishing booth model.
2. Mini game tutorials added.
3. Controls tutorial shifted
4. Voice line added, signalling end of game.
5. Failsafe features added.



Demo Video:

Link:

https://ivid2.np.edu.sg/media/GameDemoVideo_NanaStudios/1_3ezhg0vc

A demo reel of our finished game alongside live gameplay feed



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